

Epilepsy

Introduction

- Seizure:
 - ✗ transient disturbance in cerebral function caused by abnormal neuronal discharge.
- Epilepsy:
 - ✗ group of disorders represented by recurrent seizures.
 - ✗ the occurrence of at **least two unprovoked** seizures separated by **24 hours**.

Epidemiology

- Epilepsy is the fourth most common neurologic disorder.
- More than 65 million people worldwide suffer from epilepsy.
- Worldwide, the prevalence of epilepsy is believed to range from 1% to 3%.
- At least 8% of the general population will have at least one unprovoked seizure in a lifetime.
- Only 30% are in people younger than 18 years of age at the time of diagnosis.

Etiology

- 80% of patients with epilepsy, the underlying etiology is **unknown**.
- Cause of epilepsy are either from;
 - genetic mutations or
 - acquired injury (eg, stroke or traumatic brain injury)
- Epilepsy etiologies are classified into;

~~✗~~ Genetic

~~✗~~ Metabolic

~~✗~~ Structural

~~✗~~ Immune

~~✗~~ Infectious

~~✗~~ unknown

Risk Factors and Seizure Triggers

- **risk factors include;**
 - ☒ premature birth with small gestational weight
 - ☒ perinatal injury (eg, anoxia)
 - ☒ history of alcohol withdrawal seizures
 - ☒ history of febrile seizures
 - ☒ family history of seizures
- The presence of such risk factors aid in **establishing the diagnosis of epilepsy** and may help in **identifying** the underlying epilepsy etiology.

Seizure Triggers

- Physical and emotional stress
- Sleep deprivation
- Hormonal changes occurring around the time of menses
- Pregnancy
- Drugs

Pathophysiology

- Seizure results from;
 - an **excess** of excitatory neurotransmitter action (**glutamate**)
 - a **failure** of inhibitory neurotransmitter action (**GABA**), or
 - a combination of the two.
- In individual patients, it is usually **impossible to identify** which neurochemical factors are responsible.

Pathophysiology

- **Neurotransmitters (Glutamate):**

- ✎ When glutamate is released from a presynaptic neuron, it attaches to receptors on the postsynaptic neuron.

- ✎ The result is opening of membrane channels to allow sodium or calcium to flow into the postsynaptic neuron, **depolarizing** it and **transmitting** the excitatory signal.

Pathophysiology

- ✍ AED (eg, phenytoin, carbamazepine, lamotrigine) work by interfering with this mechanism;
- ✍ blocking the release of glutamate or by blocking sodium or calcium channels.
- ✍ Preventing **hyperexcitable neurons** from rapidly and **repetitively firing** and blocking firing in a use-dependent fashion.

Pathophysiology

❑ Neurotransmitters (GABA):

- ✗ It attaches to neuronal membranes and **opens chloride channels**.
- ✗ When chloride flows into the neuron, it becomes **hyperpolarized** and **less excitable**.
- ✗ This mechanism is probably critical for **suppressing seizure activity**.
- ✗ **Barbiturates** and **benzodiazepines** primarily act to enhance the action of GABA.

Seizure Classification

- Classification of epileptic seizures is based on **EEG** findings combined with the **clinical symptoms**:

1. Focal seizures

- are conceptualized as originating at some point within networks **limited to one hemisphere**

2. Generalized Seizure

- Are conceptualized as originating at some point within and rapidly engaging bilaterally distributed neural networks

a. Absence:

- **Typical absence seizures** are brief and abrupt, last 10–30 seconds, and occur in clusters.
- It usually result in a **short loss of consciousness**, or the **patient may stare, be motionless, or have a distant expression on his or her face.**
- **Electroencephalograms (EEGs)** performed during seizure activity usually **show three Hz spike-and-wave complexes.**
- are often described as **daydreaming or blanking out episodes.**
- A common term for these seizures is “**petit mal.**”
- Absence seizures **can be further classified as** typical, atypical, myoclonic absence, and eyelid myoclonia

b. Myoclonic:

- Consist of brief, lightning-like jerking movements of the entire body or the upper and occasionally lower extremities.
- Patients with these **may not lose consciousness**, due to the seizure lasting **less than 3 to 4 seconds**.
- Patients may describe these seizures as **shoulder shrugs** or **spinal chills**
- Myoclonic seizures can be further classified as
 - myoclonic,
 - myoclonic atonic, or
 - myoclonic tonic.

C.Tonic-clonic:

- ❑ Characterized by a **sudden loss of consciousness** accompanied by tonic extension and rhythmic clonic contractions of all major muscle groups.

- ❑ These seizures are often described as “**grand mal.**”

- ❑ Typically, **there are five phases** of a primary tonic-clonic seizure.....

flexion, extension, tremor, clonic, and postictal.

- **During the flexion phase,**
 - the patient's **mouth may be held partly open**, and
 - the patient may experience **upward eye movement**, involvement of the extremities, and **loss of consciousness**.
- **In the extension phase,**
 - the patient may be **noted to extend his or her back and neck**;
 - experience contraction of thoracic and abdominal muscles; be apneic; and have flexion, extension, and adduction of the extremities.
 - The patient **may cry out as air is forced from the lungs** in this phase

- The **tremor phase**
 - occurs as the patient goes from **tonic rigidity to tremors** and then to a clonic state.
- During **the clonic phase**,
 - the patient will experience **rhythmic jerks**.
- The length of the entire seizure is usually **1–3 minutes**.
 - After the seizure, the patient may be **postictal** (altered mental consciousness).
 - During this time, the patient can be difficult to arouse and/or very somnolent.

d. Clonic:

- Only the clonic phase present; rhythmic, repetitive, jerking muscle movements

e. Tonic:

- Only the flexion and/or extension phases present

f. Atonic: Characterized by a loss of muscle tone.

- Atonic seizures are often described as drop attacks in which **a patient loses tone and falls to the ground.**

3. Status Epilepticus:

- recurrent seizures of sufficient frequency that the patient does not regain consciousness between episodes.

4. Nonepileptic seizures are

- paroxysmal nonepileptic episodes resembling epileptic seizures that can be organic or psychogenic.

Clinical Presentation

- Clinical presentation of seizures varies widely depending on the **region** and **amount of brain** involved in the seizure.
 - Episodes of sudden and brief loss of consciousness
 - Episodes of uncontrolled jerking of groups of muscles
 - Sudden unexplained falls
 - Sudden and brief episodes of confusion

Diagnosis

- Thorough **clinical history**
- Complete neurological examination
- **Imaging of the brain:** MRI, CT
- **Laboratory tests:** serum electrolytes, CBC, RFT, and LFT
- **Electroencephalogram (EEG)** recording: including hyperventilation and photostimulation.

Treatment

- **Desired Outcomes:**

- ~~✎~~ elimination of seizures episode with minimal side effects of the treatment
- ~~✎~~ Improve patient QoL
- ~~✎~~ complete seizure freedom

Treatment

- **General Approach to Treatment:**

- ✎ critical to establish an accurate diagnosis of the seizure type and epilepsy classification.
- ✎ Prior to starting pharmacologic therapy, it is essential to determine the risk of having a subsequent seizure.
- ✎ If there is an underlying treatable cause, such as hyponatremia or a CNS infection, the risks of another seizure and the development of epilepsy are very small.

Treatment

- Risk factors for repeated seizures in patients without an underlying disorder include:
 - ➔ Structural CNS lesion
 - ➔ Abnormal EEG
 - ➔ Partial seizure type
 - ➔ Positive family history
 - ➔ Postictal motor paralysis
- If no risk factors are present, the risk of another seizure is 10% to 15%.
- However, if two or more risk factors are present, the risk of another seizure is 100%.

Treatment

- Despite appropriate AED treatment, approximately 30% to 35% of patients are refractory to treatment.
- The drug treatments of first choice depend on;
 - ✗ the type of epilepsy
 - ✗ patient characteristics
 - 💧 age
 - 💧 gender
 - 💧 comorbid medical conditions
 - 💧 susceptibility to adverse effects
 - 💧 ability to comply with a prescribed regimen

Mechanisms of Action of AEDs

- ✘ Act primarily by blocking the initiation or spread of seizures.
- ✘ Inhibition of Na^+ -dependent action potentials in a frequency-dependent manner (e.g., phenytoin, carbamazepine, lamotrigine, topiramate, zonisamide, lacosamide, rufinamide).
- ✘ Inhibition of voltage-gated Ca^{2+} channels (phenytoin, gabapentin, pregabalin).

Mechanisms of Action of AEDs

- ❌ Attenuation of **glutamate activity** (lamotrigine, topiramate, felbamate).
- ❌ Potentiation of **GABA receptor function** (benzodiazepines and barbiturates).
- ❌ **Increase in the availability of GABA** (valproic acid, gabapentin, tiagabine) and
- ❌ **Modulation of release of synaptic vesicles** (levetiracetam).
- ❌ **Inhibition of T-type Ca^{2+} channels** in thalamic neurons (ethosuximide and valproic acid).

When to initiate AED therapy

- If a patient presents after a single isolated seizure, one of three treatment decisions can be made:
 - treat
 - possibly treat
 - do not treat
- These decisions are based on the probability of the patient having a second seizure.
- For patients with no risk factors, normal MRI, and normal EEG, the probability of a second seizure is less than 10% - 20%.

When to initiate AED therapy

- * Two or more seizures
- * Single seizure secondary to identified CNS lesion with an epileptogenic focus
- * Consider if significant occupational risk if patient suffers a second event
- * Consider if single seizure event with two or more risk factors for recurrent seizures
- * Consider in the elderly patient with increased risk of seizure related morbidity (age, prolonged post-ictal state).

When to initiate AED therapy

- **To start AED therapy:**

- 👉 monotherapy is preferred.

- 👉 approximately 50% to 70% of all patients with epilepsy can be maintained on one drug.

- 👉 The percentage of patients who are seizure free on one drug varies by seizure type.

- 👉 Combining AEDs with different mechanisms of action to achieve freedom from seizures of this approach is yet **unproven**.

To start AED therapy

- AED therapy should usually be initiated carefully using a titration schedule to minimize adverse events.
- Increased the dose gradually until the patient becomes seizure free or adverse effects appear.
- Combinations of drugs may be useful in patients with difficult to control seizures.
- Those who have unsatisfactory control despite multiple drug treatment may be candidates for vagal nerve stimulators, surgery

Switching therapy

- Changing from one AED to another can be a complex process.
- If the first drug is stopped too abruptly, breakthrough seizures may occur.
- Typically the new drug is started at a low initial dose and gradually increased over several weeks.
- Once the new drug is at a **minimally effective dose**, the drug to be discontinued is gradually tapered while the dose of the new drug continues to be **increased to the target dose**.

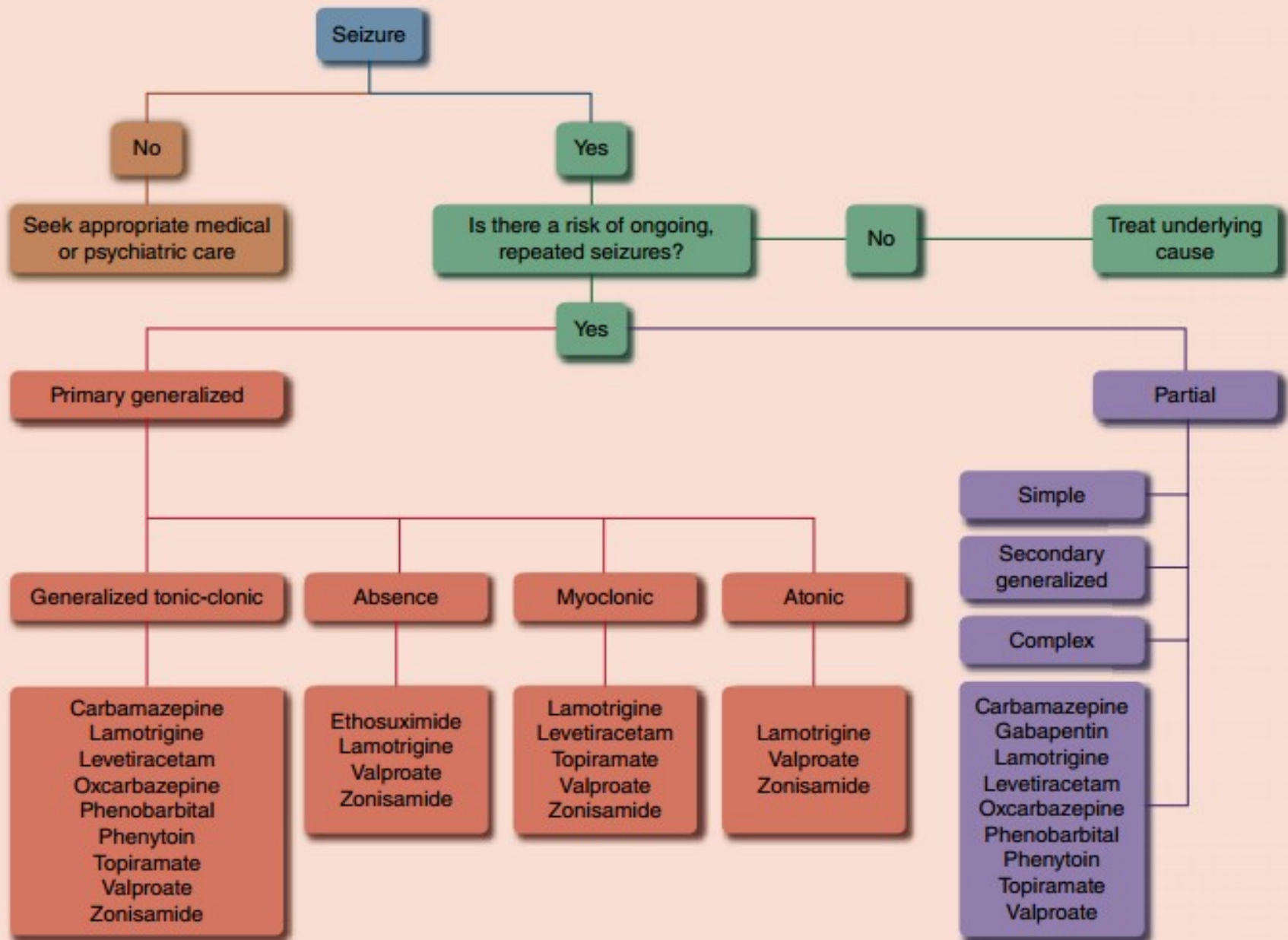
Stopping therapy

- The AEDs used to control seizures may not need to be given for a lifetime.
- **Five criteria must be met** before considering the discontinuation of AED;
 - ➔ No seizures for 2 to 5 years
 - ➔ Normal neurologic examination
 - ➔ Normal intelligence quotient
 - ➔ Single type of partial or generalized seizure
 - ➔ Normal EEG with treatment

Stopping therapy

- Individuals who fulfill all of these criteria have a **65% chance of remaining seizure-free after AEDs are discontinued.**
- AEDs can be restarted in patients who relapse after AED withdrawal.
- **Withdrawal of AEDs is done slowly, usually with a tapering dose over at least 3 months.**

Treatment algorithm for seizures



Adverse effect

Carbamazepine:

- Rash (occurs after a delay of 2–8 weeks),
- syndrome of inappropriate antidiuretic hormone release,
- Aplastic anemia, thrombocytopenia, anemia, leukopenia
- Patients with the *HLA-B*1502* allele are
 - at a 10 times increased risk of Stevens-Johnson syndrome.
- Testing is recommended for
 - Asians (including Indians).
- More than 15% of populations in
 - Hong Kong, Malaysia, the Philippines, and Thailand have this allele.

Adverse effect...

Phenytoin:

- Dose-related adverse effects:
 - Nystagmus, ataxia, drowsiness, cognitive impairment
- Non—dose-related adverse effects:
 - Gingival hyperplasia, hirsutism, acne, rash,
 - hepatotoxicity, coarsening of facial features

ADR...

Valproic Acid:

- Hepatotoxicity,
- nausea/vomiting,
- weight gain,
- interference with platelet aggregation,
- pancreatitis,
- alopecia,
- tremor

ADR...

Phenobarbitone:

- Hyperactivity,
- cognitive impairment
- Fatigue
- drowsiness,
- depression

ADR...

Sexual dysfunction

- Described in
 - 30%–60% of men and women with epilepsy
- Includes hyposexuality, orgasmic dysfunction, and erectile dysfunction
- Sexual dysfunction has been reported with
 - carbamazepine, phenobarbital, phenytoin, pregabalin, topiramate, and zonisamide.
- **Improved sexual functioning** has been reported with lamotrigine and oxcarbazepine.

ADR...

Osteopenia or osteoporosis

- found in 38%–60% of patients in tertiary epilepsy clinics.
- Increased fractures in patients with epilepsy and with seizure medication use
- Risk is increased with
 - increased treatment duration;
 - there is a dose-response relationship;

ADR- Bone health

- The medications most often associated with poor bone health are:
 - carbamazepine, clonazepam, phenobarbital, and valproic acid.
 - However, there is **now evidence that all seizure medications** may contribute to osteopenia or osteoporosis.
- Proposed mechanisms:
 - Hepatic induction of CYP isoenzymes leads to **increased vitamin D catabolism; impaired calcium absorption;**
 - calcitonin deficiency; vitamin K interference; and
 - direct detrimental effect on bone cells.

ADR- Bone health

- Proposed treatments:
 - **Vitamin E**..... High-dose (4000 international units/day for adults and
 - 2000 international units/day for children)
 - **Vitamin D** improves bone mineral density
 - **Estrogen** may be helpful for women, but may also trigger seizures in some women

Drug Interactions

- AEDs are associated with many different drug interactions.
- Most interactions occur due to alterations in absorption, metabolism, and protein binding.
- Tube feedings and antacids are known to **reduce the absorption** of phenytoin and carbamazepine.
- Phenytoin, carbamazepine, and phenobarbital are **potent inducers of various CYP450** isoenzymes, increasing the clearance of other drugs metabolized through these pathways.
- Valproate is a CYP450 isoenzyme **inhibitor** and reduces the clearance of some drugs.

Drug Interactions

- Phenytoin and valproate **are highly protein-bound** and can be displaced when taken together or with other highly protein-bound drugs.
- When phenytoin and valproate are taken together, there may be increased dose-related adverse effects within several hours of dosing.
- This can be avoided by staggering doses or giving smaller doses more frequently during the day.
- Whenever a change in a medication regimen occurs, drug interactions should be considered and appropriate adjustments in dose of AEDs made.

Drug Interactions

Cytochrome P450 and AED Interactions

Enzyme	Substrate	Common Inducers	Common Inhibitors
CYP 1A2	Carbamazepine	Carbamazepine Phenytoin Phenobarbital Rifampin	Cimetidine Ciprofloxacin Erythromycin Clarithromycin
CYP 2C9	Phenobarbital ^a Phenytoin ^a Carbamazepine Valproate	Carbamazepine Phenytoin Phenobarbital Rifampin	Amiodarone Cimetidine Fluconazole Valproate
CYP 2C19	Phenobarbital Phenytoin Valproate Lacosamide		Felbamate Ticlopidine Topiramate Zonisamide
CYP 2D6 CYP 3A4	Zonisamide Carbamazepine ^a Tiagabine ^a Zonisamide ^a	Carbamazepine Carbamazepine Phenytoin Phenobarbital Rifampin	Amiodarone Erythromycin Ketoconazole
Uridine diphosphate glucuronyl-transferase	Lamotrigine ^a Carbamazepine Valproate	Lamotrigine Phenobarbital Phenytoin Hormonal contraceptives	Valproate

Comparison of older and newer AEDs

Area of Comparison	AED Group	
	Traditional AEDs [*]	Newer AEDs [†]
Efficacy	Well established	Equally good (probably), but less well established
Clinical experience	Extensive	Less extensive
Therapeutic niche	Well established	Evolving
Tolerability	Less well tolerated	Better tolerated (usually)
Pharmacokinetics	Often complex	Less complex
Drug interactions	Extensive, owing to induction of drug-metabolizing enzymes	Limited, owing to little or no induction of drug-metabolizing enzymes
Safety in pregnancy	Less safe	Safer
Cost	Less expensive	More expensive

^{*}Carbamazepine, ethosuximide, phenobarbital, fosphenytoin, phenytoin, primidone, and valproic acid.

[†]Ezogabine, felbamate, gabapentin, lacosamide, lamotrigine, levetiracetam, oxcarbazepine, pregabalin, rufinamide, tiagabine, topiramate, vigabatrin, and zonisamide.

Special Populations

- **Children and women with epilepsy** have unique problems related to the use of AEDs.
- **In children:**
 - ✗ developmental changes occur rapidly, and metabolic rates are greater than those seen in adults.
 - ✗ it is imperative to control seizures as quickly as possible to avoid interference with development of the brain and cognition.
 - ✗ AED doses are **increased rapidly**, and frequent changes in the regimen are made to maximize seizure control.

Special Populations

- **For women:**

- ✗ the treatment of epilepsy poses challenges, including **teratogenicity, breastfeeding**, interactions between AEDs and **hormonal contraceptives**, and **reduced fertility**.
- ✗ Several AEDs are implicated in causing birth defects.
 - ✗ **Neural tube defects** (eg, spina bifida, microcephaly, anencephaly)
 - ✗ **Impaired cognitive development** in children born to women

Pregnancy

- **Avoiding**
 - valproic acid monotherapy or polytherapy during the first trimester of pregnancy
 - This is because of the risk of major congenital malformations, particularly **neural tube defects**, facial clefts, hypospadias, and poor cognitive outcomes.
- **Avoiding the use of.....**
 - **phenytoin, carbamazepine, and phenobarbital**, if possible;
 - cleft palate (phenytoin), posterior cleft palate (carbamazepine), cardiac malformations (phenobarbital), and poor cognitive outcomes (phenytoin, phenobarbital).

Management of AEDs During Pregnancy

- Give supplemental folic acid 1–4 mg daily to all women of childbearing potential
- Use monotherapy whenever possible
- Use lowest doses that control seizures
- Continue pharmacotherapy that best controls seizures prior to pregnancy
- Monitor AED serum concentrations at start of pregnancy and monthly thereafter
- Adjust AED doses to maintain baseline serum concentrations
- Administer supplemental vitamin K during eighth month of pregnancy to women receiving enzyme-inducing AEDs
- Monitor postpartum AED serum concentrations to guide adjustments of drug doses

Outcome Evaluation

- Patients receiving AEDs for seizures should have regular monitoring for **seizure frequency**, **seizure patterns**, **acute & chronic** adverse effects, **comorbid conditions**, and possible **drug interactions**.
- **Efficacy:**
 - Seizure counts are the standard way to evaluate the efficacy of treatment.
 - Encourage patients to keep a seizure calendar that notes the time and day a seizure occurs and the type of seizure.
- **Toxicity:**
 - Monitor toxicity of AEDs at every clinic visit.

Nonpharmacologic Therapy

- Nonpharmacologic therapy for epilepsy includes;
 - Ketogenic diet
 - Surgery
 - Vagus nerve stimulation (VNS):
 - ✍ mechanisms of antiseizure actions of VNS are unknown
 - ✍ reducing the frequency of seizures with partial-onset seizures that are refractory to AEDs
 - ✍ It is also used off-label in the treatment of refractory primary generalized epilepsy

Nonpharmacologic Therapy

- **Surgery:**

- ☞ the treatment of choice in selected patients with refractory focal epilepsy, especially those patients with seizures originating from the **temporal lobe**.

- ☞ for patient refractory to combination drug.

- Role of **ketogenic diet**???????

Status Epilepticus (SE)

Introduction

- SE:

👉 is continuous seizure activity **lasting more than 5 minutes** or **two or more seizures without complete recovery of consciousness.**

👉 is a neurologic emergency that can lead to **permanent brain damage or death.**

- **Refractory status epilepticus (RSE)** is unresponsive to emergent (first-line) or urgent (second-line) therapy.

Introduction

- SE can present as;
 - ➔ non-convulsive status epilepticus (NCSE)
 - ➔ generalized convulsive status epilepticus (GCSE)
- **NCSE:**
 - ✗ x/zed by **persistent** impaired consciousness **without clinical seizure** activity and is diagnosed with EEG.
- **GCSE:**
 - ✗ x/zed by full-body motor seizures and involves the entire brain.

Epidemiology

- The incidence of SE in the US is 12.5 / 100,000 / year.
- In-hospital mortality rate of 9.2%.
- Most SE occurs in individuals with **no history of epilepsy**.
- Approximately 5% of adults and 10% to 25% of children with epilepsy will develop SE.
- The incidence is highest in those younger than 1 year of age and in those older than 60 years of age.

Etiology

- **Causes of SE include:**
 - ☞ metabolic disturbances
 - ☞ CNS disorders
 - ☞ hypoxia
 - ☞ Drug toxicity (eg, theophylline, isoniazid, cyclosporine, cocaine)

Etiology

❖ **Chronic causes of SE include;**

- ☞ preexisting epilepsy
- ☞ chronic alcoholism (withdrawal seizures)
- ☞ CNS tumors
- ☞ strokes

☑ **In epileptics, the common causes of SE are anticonvulsant withdrawal or subtherapeutic levels.**

Pathophysiology

- Status epilepticus occurs when the **brain fails to stop isolated seizures**, due to a **mismatch of neurotransmitters**.
- Glutamate, stimulates postsynaptic *N*-methyl-d-aspartate (NMDA) receptors, causing **neuronal depolarization**.
- Sustained depolarization causes **neuronal injury** and **death**.
- GABA, **opposes excitation** by stimulating **GABA_A receptors**, producing **hyperpolarization** and **inhibition of the postsynaptic cell membrane**.

Clinical Presentation

- **General:**

- ✗ The patient may present with or without clinically noticeable seizure activity.

- **Symptoms:**

- ✗ Impaired consciousness ranging from lethargy to coma
- ✗ Disorientation after cessation
- ✗ Pain from associated injuries (eg, tongue lacerations, dislocated shoulder, head trauma, facial trauma).

Clinical Presentation

Signs:

• Phase I:

- ✗ Occurs during the first 30 minutes
- ✗ Generalized convulsions
- ✗ Hypertension, tachycardia
- ✗ Fever and sweating
- ✗ Muscle contractions, spasms
- ✗ Respiratory compromise
- ✗ Incontinence

• Phase II:

- ✗ greater than 30 minutes of SE
- ✗ Respiratory failure with pulmonary edema
- ✗ Cardiac failure (arrhythmias, shock)
- ✗ Hypotension
- ✗ Hyperthermia
- ✗ Rhabdomyolysis and multiorgan failure

Diagnosis

Laboratory Tests

- Hyperglycemia (phase I) and hypoglycemia (phase II) can occur
- Hyponatremia, hypernatremia, hyperkalemia, hypocalcemia, hypomagnesemia, and hypoglycemia can cause SE
- The white blood cell (WBC) count may slightly increase
- Abnormal arterial blood gases (ABGs) due to hypoxia and respiratory or metabolic acidosis
- Elevated serum creatinine will be present in renal failure patients
- Myoglobinuria can occur in patients with continuous seizures

Diagnostic Tests

EEG to confirm seizure activity

CT or MRI may reveal mass lesions or hemorrhage

Treatment

- **Desired Outcomes:**

- ☒ **Short-term desired outcomes:**

- immediate termination of all clinical and electrical seizure activity
 - minimize clinically significant adverse effects
 - prevent recurrent seizure activity

- ☒ **The long-term outcomes:**

- minimizing or avoiding pharmacoresistant epilepsy
 - reducing development of neurologic sequelae

Pharmacologic Therapy

- **There are four immediate goals:**
 - ✗ patient stabilization, including adequate oxygenation, preservation of cardiorespiratory function, and management of systemic complications
 - ✗ accurate diagnosis of the subtype of GCSE and identification of **precipitating factors**
 - ✗ termination of clinical and electrical seizures as early as possible
 - ✗ **prevention** of seizure recurrence
- The **benzodiazepines, hydantoins, and barbiturates** are the most commonly used for the initial treatment of GCSE.

Emergency Medications

Lorazepam: Drug of choice

- Rapid onset (2–3 minutes)
- Dosage 0.1 mg/kg (up to 4 mg/dose) at rate of up to 2 mg/minute; may repeat every 5–10 minutes

Diazepam

- Rapid onset
- Dosage 0.15 mg/kg (up to 10 mg/dose) at rate of up to 5 mg/minute. May repeat every 5 minutes
- **Rectal gel formulation** can be given in absence of intravenous access.

Emergency medicines

Phenytoin:

- Dosage 20 mg/kg;
- administration rate less than 50 mg/minute

Fosphenytoin:

- Administration rate less than 150 mg of phenytoin equivalent per minute

Phenobarbital:

- Dosage 20 mg/kg at 50–100 mg/minute

Refractory SE

Pentobarbital:

- **Load** 5–15 mg/kg up to 50 mg/minute;
- follow with a 0.5- to 5-mg/kg/hour infusion
- May have severe hypotension, requiring treatment with vasopressors;
- should have continuous BP measurement
- **Must be on ventilator**

Refractory SE

Thiopental:

- Load 2–7 mg/kg up to 50 mg/minute;
- follow with 0.5- to 5-mg/kg/hour infusion.
- (a) May have severe hypotension, respiratory depression, cardiac depression
- (b) Must be on ventilator

Midazolam:

- Load 0.2-mg/kg infused up to 2 mg/minute;
- follow with a 0.05- to 2-mg/kg/hour infusion.
- (a) May have hypotension, respiratory depression
- (b) May experience tachyphylaxis

Refractory SE

Propofol:

- Load a 1- to 2-mg/kg intravenous bolus for 30–60 seconds;
- follow with a 20- to 200- mcg/kg/minute infusion.
- (a) Significant source of lipids
- (b) Some reports of seizure exacerbation with propofol
- (c) Must be on ventilator

Algorithm for the treatment of GCSE

PREHOSPITAL CARE

- Monitor Vital signs (HR, RR)
- PR diazepam (0.2-0.5 mg/kg)
- Consider IN midazolam (0.15-0.3 mg/kg) or IM midazolam (0.2 mg/kg up to 10 mg)
- Transport to hospital if seizures persist

INITIAL HOSPITAL CARE

- Time from seizure onset
- Assess and control airway and cardiac function; pulse oximetry
- 100% oxygen
- Place IV catheter; Intraosseous if unable to place IV and patient is younger than 6 years
- Begin IV fluids; blood pressure support as needed
- Thiamine 100 mg (adult)
- Pyridoxine 50-100 mg (infant)
- Glucose (adult: 50 mL of 50%; children: 1 mL/kg of 10%) if serum glucose is <60 mg/dL (<3.3 mmol/L)
- Naloxone 0.1 mg/kg for suspected narcotic overdose
- Antibiotics if infection suspected
- Treat hyperthermia

LABORATORY STUDIES

- CBC with differential
- Serum chemistry profile (eg, electrolytes, glucose, renal/hepatic function, calcium, magnesium)
- Arterial blood gas
- Blood cultures
- Serum anticonvulsant concentration
- Urine drug/alcohol screen
- EEG as needed

IMPENDING GSCE (0-30 minutes)

- IV lorazepam (0.1 mg/kg over 30-60 seconds up to 6 mg) may repeat in 5 minutes if no response or IN midazolam (0.15-0.3 mg/kg) or IM midazolam (0.2 mg/kg up to 10 mg)
- Additional therapies may not be required if seizure stops

ESTABLISHED GSCE (30-60 minutes)

First-line

Phenytoin (IV): 18-20 mg/kg over 20 minutes (<1 mg/kg/min; max 50 mg/min); may give an additional 5 mg/kg as needed or Fosphenytoin (IV or IM): 18-20 mg PE/kg (<3 mg PE/kg/min; max <150 mg PE/min)

Second-line

Employment, Driving, and Other Activities

❖ Driving privileges

- Most states allow patients to drive after a seizure-free interval (on or off medications) of between 3 months and 2 years.

❖ Working at heights or with machinery

❖ Bathing and swimming